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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

LOS -Theory 6 Assignment

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MANIPAL SCHOOL OF INFORMATION SCIENCES
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Assignment: Cybersecurity Log Analysis Using Linux Tools

Title: Detection and Analysis of SSH Brute-Force Login Attempts Using Linux Command-Line Tools

1. Introduction

Cybersecurity log analysis is the process of examining system-generated log files to identify suspicious activities, security threats, unauthorized access attempts, and abnormal system behaviour.

Linux systems maintain different types of log files that record information about user authentication, system activities, network connections, and security events.

In this assignment, Linux command-line tools are used to analyse authentication logs and identify suspicious SSH login attempts. The analysis combines power filters, Bash scripting, grep, sed, awk, sort, uniq, and pipelines.

The main objective is to develop a Bash script that automatically processes authentication logs, counts failed login attempts, identifies suspicious IP addresses, and generates an analysis report.

2. Selected Cybersecurity Scenario

SSH Brute-Force Login Attack Detection

An SSH (Secure Shell) brute-force attack is a cybersecurity attack in which an attacker repeatedly attempts to log in to a remote Linux system by trying different usernames and passwords.

The attacker may attempt to gain unauthorized access by guessing valid login credentials.

For example, an attacker might repeatedly attempt to log in using common usernames such as:

- root
- admin
- test
- guest
- user

If the attacker successfully guesses valid credentials, they may gain unauthorized access to the system.

Security Significance

Detecting SSH brute-force attacks is important because:

1. It helps identify repeated unauthorized login attempts.
2. It helps administrators detect suspicious IP addresses.
3. It helps identify attempts to access privileged accounts.
4. It helps prevent unauthorized access to Linux systems.
5. It supports the early detection of possible security incidents.

In this assignment, a simulated SSH authentication log is analysed to identify repeated failed login attempts from the same IP address.

3. Method Used to Obtain the Log File

A safe and authorized simulation is used to obtain the authentication log data.

Simulation Environment

Component	Description
Operating System	Kali Linux
Log Type	SSH Authentication Log
Log Format	Linux authentication log format
Tools	Bash, grep, sed, awk, sort, uniq, cut
Scenario	Simulated SSH brute-force login attempts
Target	Locally created sample log file

Procedure

Instead of generating attacks against an external system, a synthetic authentication log file is manually created using sample SSH login records.

The file contains:

- Failed SSH login attempts.

- Invalid username attempts.
- Successful SSH login attempts.
- Authentication attempts from different IP addresses.
- Normal system authentication activities.

This approach provides a safe environment for practising cybersecurity log analysis.

Dataset Source: Personally created synthetic log file.

Source URL: Not applicable.

Access Date: 06 November 2026 (date of creation).

4. Structure of the Log File

The authentication log contains information about different authentication events.

A typical SSH authentication log entry looks like:

Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2

Important Fields

Field	Description
Nov 06	Date of the event
10:00:01	Time of the event
kali	Hostname of the system
sshd	SSH daemon service
2101	Process ID
Failed password	Authentication result
admin	Username attempted
192.168.56.101	Source IP address
50111	Source port number
ssh2	SSH protocol version

Important Log Events

Failed password: Indicates that an authentication attempt was unsuccessful.

Accepted password: Indicates that a successful password-based authentication occurred.

Invalid user: Indicates that an attempted username does not correspond to a valid system account.

sshd: Represents the SSH daemon responsible for handling SSH connections.

5. Cybersecurity Investigation Questions

The following investigation questions are formulated for analysing the authentication log.

Question 1: Which IP addresses generated the highest number of failed SSH login attempts?

Question 2: Are there any IP addresses that attempted to log in using multiple invalid usernames?

Question 3: Are there repeated failed authentication attempts that may indicate a brute-force attack?

Question 4: Were there any successful login attempts in the authentication log?

Question 5: Which IP addresses should be investigated further based on the number of failed login attempts?

6. Creation of the Log File

First, create a project directory.

mkdir -p ~/cyber-log-analysis

Navigate into the directory.

cd ~/cyber-log-analysis

Create a sample authentication log file.

nano auth.log

Enter the following synthetic log records:

Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2

Nov 06 10:00:08 kali sshd[2102]: Failed password for root from 192.168.56.101 port 50112 ssh2

Nov 06 10:00:15 kali sshd[2103]: Failed password for invalid user test from 192.168.56.101 port 50113 ssh2

Nov 06 10:00:22 kali sshd[2104]: Failed password for invalid user guest from 192.168.56.101 port 50114 ssh2

Nov 06 10:00:29 kali sshd[2105]: Failed password for invalid user oracle from 192.168.56.101 port 50115 ssh2

Nov 06 10:00:36 kali sshd[2106]: Failed password for invalid user admin from 192.168.56.101 port 50116 ssh2

Nov 06 10:01:01 kali sshd[2107]: Failed password for invalid user test from 192.168.56.102 port 50201 ssh2

Nov 06 10:01:15 kali sshd[2108]: Failed password for root from 192.168.56.102 port 50202 ssh2

Nov 06 10:02:01 kali sshd[2109]: Accepted password for kali from 192.168.56.103 port 50301 ssh2

Nov 06 10:03:01 kali sshd[2110]: Failed password for invalid user admin from 192.168.56.104 port 50401 ssh2

Nov 06 10:04:01 kali sshd[2111]: Accepted password for kali from 192.168.56.103 port 50302 ssh2

Nov 06 10:05:01 kali sshd[2112]: Failed password for root from 192.168.56.104 port 50402 ssh2

Save the file using CTRL + O, press Enter, and exit using CTRL + X.

Verify the file:

cat auth.log

7. Initial Log Analysis Using Linux Power Filters

Before developing the Bash script, individual Linux commands are used to examine the authentication log.

7.1 Display the Log File

cat auth.log

Purpose: Displays the complete contents of the authentication log.

7.2 Count Total Log Records

wc -l auth.log

Purpose: Counts the total number of lines in the log file.

Expected output:

12 auth.log

7.3 Find Failed Login Attempts

grep "Failed password" auth.log

Purpose: Searches for authentication records containing failed password attempts.

7.4 Find Successful Login Attempts

grep "Accepted password" auth.log

Purpose: Identifies successful password-based SSH authentication events.

7.5 Extract Source IP Addresses

grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}'

Purpose: Extracts the IP addresses associated with failed login attempts.

7.6 Count Failed Attempts from Each IP Address

grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort | uniq -c

Purpose:

- grep identifies failed login records.
- grep -oE extracts IP addresses.
- sort arranges identical IP addresses together.
- uniq -c counts the occurrences of each IP address.

Expected output:

6 192.168.56.101

2 192.168.56.102

2 192.168.56.104

7.7 Display the Most Frequent IP Addresses

grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort | uniq -c | sort -nr

Purpose: Sorts the IP addresses according to the number of failed attempts in descending order.

7.8 Display the First Five Log Records

head -n 5 auth.log

Purpose: Displays the first five records.

7.9 Display the Last Five Log Records

tail -n 5 auth.log

Purpose: Displays the last five records.

8. Analysis Using sed

The sed command is a stream editor used to search, filter, and transform text.

In this assignment, sed is used to extract only failed SSH authentication records.

Create a separate sed script:

nano failed_logins.sed

Enter:

/Failed password/p

Save the file.

Execute the sed script:

sed -n -f failed_logins.sed auth.log

Purpose:

- sed processes the authentication log.
- /Failed password/ searches for failed authentication records.

- p prints matching records.
- -n prevents automatic printing of all records.
- -f loads the sed commands from the specified script file.

The output contains only failed SSH login attempts.

9. Analysis Using awk

The awk command is used for pattern scanning, field extraction, counting, and data processing.

Create an AWK script:

nano analyze.awk

Enter the following code:

```
{
   ip = ""

   for (i = 1; i <= NF; i++) {
      if ($i == "from") {
         ip = $(i + 1)
      }
   }

   if (ip != "") {
      count[ip]++
   }
}

END {
   print "FAILED LOGIN ANALYSIS"
```

```
    print "---------------------"

    for (ip in count) {
        status = (count[ip] >= 5) ? "SUSPICIOUS" : "NORMAL"

        printf "%-18s %d attempts  %s\n", ip, count[ip], status
    }
}
```

Save the file.

Execute:

sed -n -f failed_logins.sed auth.log | awk -f analyze.awk

Purpose:

- NF represents the total number of fields in a record.
- $i accesses individual fields.
- The for loop searches for the word from.
- $(i+1) extracts the IP address immediately following from.
- count[ip]++ increases the failed login count for each IP address.
- END executes after all records have been processed.
- printf displays the IP address, attempt count, and classification.
- The condition count[ip] >= 5 identifies potentially suspicious IP addresses.

10. Development of the Bash Script

A Bash script is developed to combine multiple Linux tools and automate the complete log analysis process.

The script performs the following operations:

1. Checks whether the input log file exists.
2. Counts total log records.

3. Extracts failed authentication attempts using sed.
4. Extracts successful authentication attempts.
5. Counts failed attempts from individual IP addresses.
6. Uses awk to classify suspicious IP addresses.
7. Generates an analysis report.
8. Displays the final results.

Create the Bash Script

nano analyze_logs.sh

Enter the following code:

```
#!/bin/bash

LOG="auth.log"
OUTPUT="output"

if [ ! -f "$LOG" ]; then
   echo "Error: Log file not found!"
   exit 1
fi

mkdir -p "$OUTPUT"

echo "CYBERSECURITY LOG ANALYSIS"
echo "============================"

TOTAL=$(wc -l < "$LOG")
```

```
echo "Total log records: $TOTAL"

sed -n -f failed_logins.sed "$LOG" > "$OUTPUT/failed.log"

grep "Accepted password" "$LOG" > "$OUTPUT/successful.log"

FAILED=$(wc -l < "$OUTPUT/failed.log")
SUCCESS=$(wc -l < "$OUTPUT/successful.log")

echo "Failed login attempts: $FAILED"
echo "Successful login attempts: $SUCCESS"

echo
echo "FAILED ATTEMPTS BY IP"
echo "---------------------"

grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' "$OUTPUT/failed.log" |
sort | uniq -c | sort -nr |
tee "$OUTPUT/ip_counts.txt"

echo
echo "SUSPICIOUS IP ANALYSIS"
echo "----------------------"

awk -f analyze.awk "$OUTPUT/failed.log" |
tee "$OUTPUT/analysis.txt"
```

```
echo
echo "SUCCESSFUL LOGIN RECORDS"
echo "----------------------"

cat "$OUTPUT/successful.log"

echo
echo "Analysis completed."
echo "Reports saved in the output directory."
```

Save the file.

Make the Script Executable

```
chmod +x analyze_logs.sh
```

Execute the Script

```
./analyze_logs.sh
```

11. Explanation of Commands Used in the Bash Script

Command	Purpose
#!/bin/bash	Specifies Bash as the script interpreter.
if [! -f "$LOG"]	Checks whether the log file exists.
mkdir -p	Creates the output directory if it does not exist.
wc -l	Counts the total number of log records.
sed -n -f	Extracts failed login records using the separate sed script.
grep	Searches for successful authentication records.
grep -oE	Extracts IP addresses using an extended regular expression.
sort	Sorts extracted IP addresses.

uniq -c	Counts repeated IP addresses.
sort -nr	Sorts numerical counts in descending order.
awk -f	Executes the separate AWK analysis script.
tee	Displays output and saves it into a file.
cat	Displays successful login records.
chmod +x	Gives executable permission to the Bash script.

12. Output and Findings

After executing the Bash script, the following results are obtained.

Total Log Records

Total log records: 12

Failed login attempts: 10

Successful login attempts: 2

Failed Attempts by IP Address

6 192.168.56.101

2 192.168.56.102

2 192.168.56.104

Suspicious IP Analysis

FAILED LOGIN ANALYSIS

192.168.56.101 6 attempts SUSPICIOUS

192.168.56.102 2 attempts NORMAL

192.168.56.104 2 attempts NORMAL

The order of IP addresses displayed by AWK may vary because AWK does not guarantee associative-array iteration order.

Successful Login Records

Nov 06 10:02:01 kali sshd[2109]: Accepted password for kali from 192.168.56.103 port 50301 ssh2

Nov 06 10:04:01 kali sshd[2111]: Accepted password for kali from 192.168.56.103 port 50302 ssh2

Generated Output Files

The Bash script creates the following output directory:

cyber-log-analysis/

├── auth.log

├── analyze_logs.sh

├── failed_logins.sed

├── analyze.awk

└── output/

  ├── failed.log

  ├── successful.log

  ├── ip_counts.txt

  └── analysis.txt

13. Identification of Suspicious Activities

The analysis identifies three IP addresses that generated failed SSH authentication attempts.

Suspicious IP Address: 192.168.56.101

- Total failed attempts: 6.
- Attempted usernames include admin, root, test, guest, and oracle.
- Repeated authentication failures occurred within a short period.
- Multiple invalid usernames were attempted.

- The activity is consistent with a possible brute-force or username-guessing attempt.

IP Address: 192.168.56.102

- Total failed attempts: 2.
- Attempted usernames include test and root.
- The number of failures is below the selected threshold.
- Further investigation may be necessary if similar activity continues.

IP Address: 192.168.56.104

- Total failed attempts: 2.
- Attempted usernames include admin and root.
- The number of attempts is below the suspicious activity threshold.

Successful Login Activity

The IP address 192.168.56.103 generated two successful authentication events.

These events are not automatically classified as malicious because successful authentication does not necessarily indicate unauthorized access. The identity of the user and the legitimacy of the connection would need to be verified.

14. Justification of Suspicious Activity Threshold

For this assignment, an IP address is classified as suspicious when it generates five or more failed SSH login attempts in the analysed log file.

The threshold is selected to demonstrate how repeated authentication failures can be used to identify potential brute-force activity.

Number of Failed Attempts	Classification
1–4	Normal / Monitor
5 or more	Suspicious

The IP address 192.168.56.101 generated six failed attempts and therefore satisfies the selected suspicious activity condition.

This threshold is an educational rule for the simulated dataset. It is not a universal cybersecurity standard. In a real environment, the threshold should also consider the time window, user behaviour, account importance, and other security indicators.

15. Possible False Positives

False positives occur when legitimate activities are incorrectly classified as suspicious.

Possible reasons include:

1. A legitimate user repeatedly entering an incorrect password.
2. An administrator testing multiple user accounts.
3. An automated application using outdated credentials.
4. A user accidentally attempting to access the wrong SSH server.
5. Multiple users accessing a system through a shared IP address.

Therefore, repeated failed login attempts should be investigated before concluding that an attack has occurred.

16. Limitations of the Analysis

The developed analysis has certain limitations:

1. The assignment uses a synthetic log file rather than real production authentication logs.
2. The analysis is based mainly on failed login counts.
3. The script does not verify whether a source IP address is malicious.
4. The analysis does not automatically block suspicious IP addresses.
5. The selected threshold may not be suitable for every Linux environment.
6. The script does not correlate authentication events with firewall or network traffic logs.
7. The script does not perform advanced behavioural analysis or machine learning.

Despite these limitations, the script demonstrates how basic Linux tools can be combined to perform useful preliminary security investigations.

This assignment demonstrates the use of Linux command-line tools for cybersecurity log analysis.

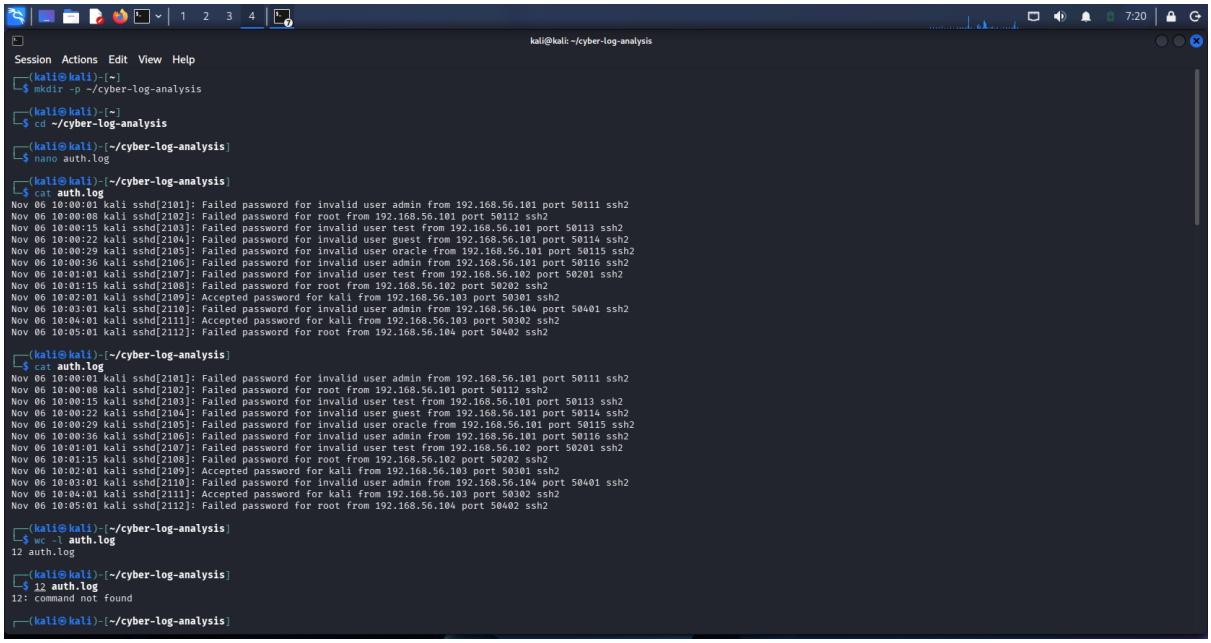
A simulated SSH authentication log was created and analysed to identify possible brute-force login attempts.

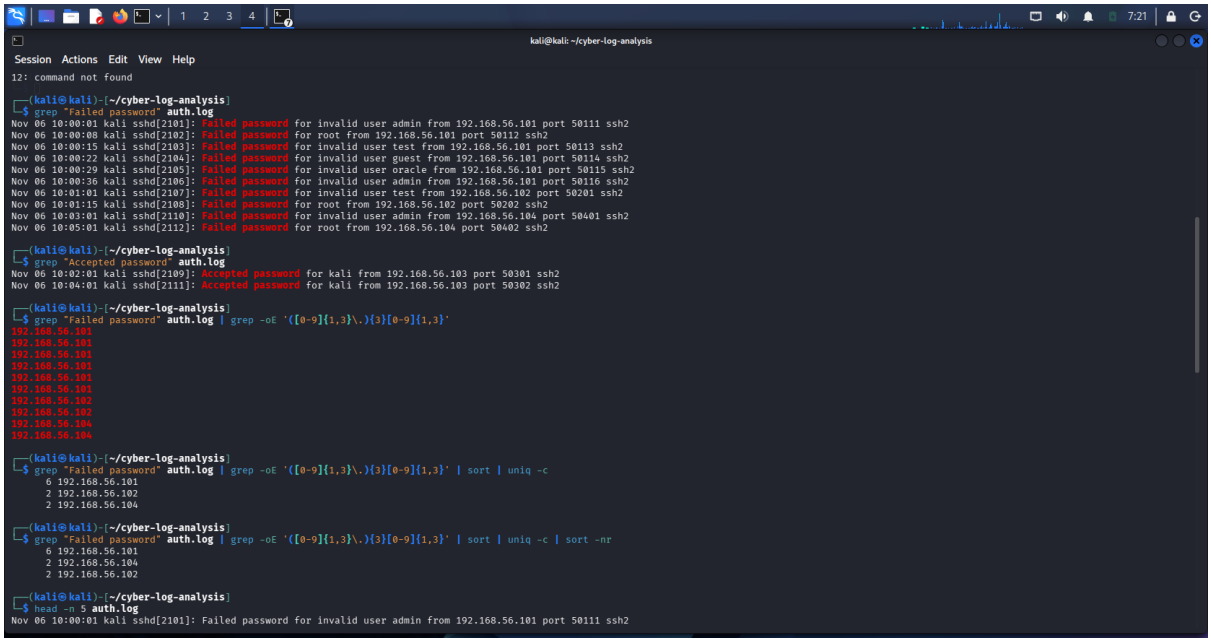
Different Linux tools were combined to perform the analysis. grep was used to identify authentication events, sed was used to filter failed login records, and awk was used to extract IP addresses, count attempts, and classify suspicious activities. Additional commands such as sort, uniq, wc, cut, head, and tail demonstrate the use of Linux power filters and pipelines.

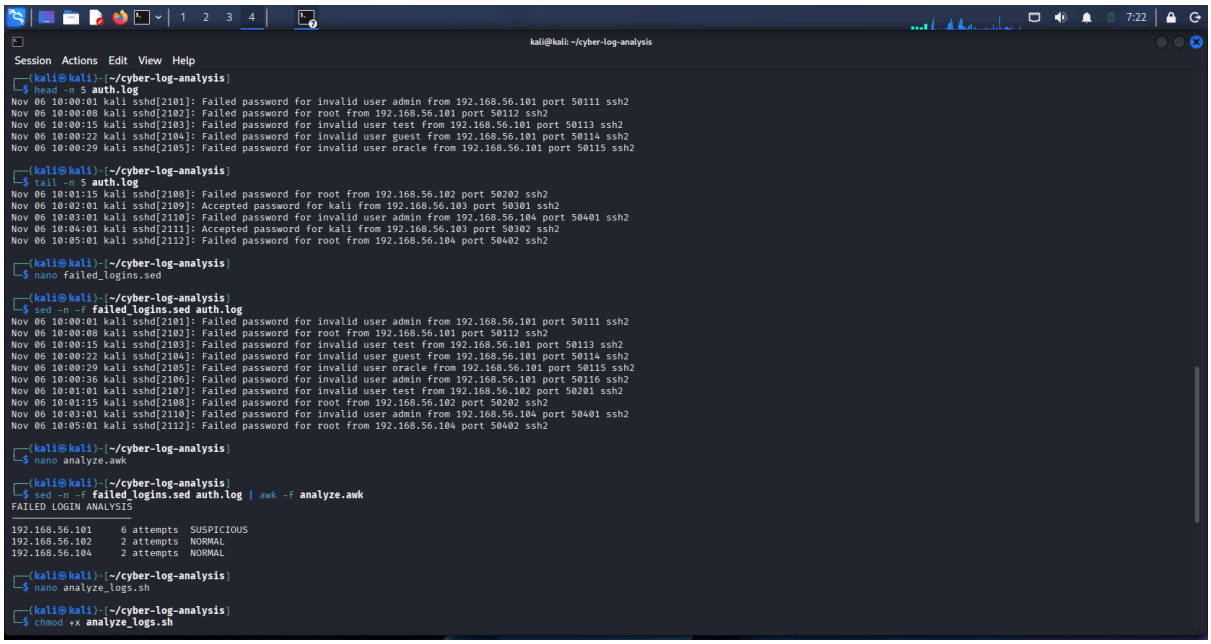
The Bash script integrates these tools and automatically generates analysis results and output files.

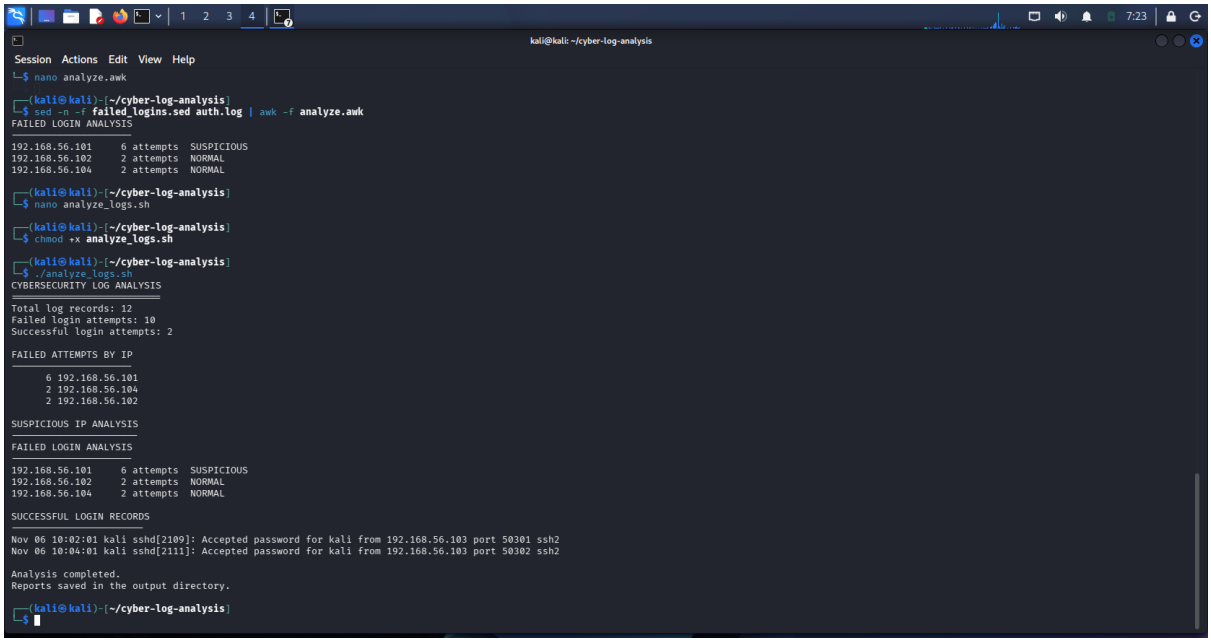
The analysis identified 192.168.56.101 as a suspicious IP address because it generated six failed SSH authentication attempts and attempted multiple invalid usernames.

This assignment helped demonstrate how Linux tools and Bash scripting can be used for basic security monitoring, log investigation, and preliminary detection of suspicious authentication activities.




kali@kali: ~/cyber-log-analysis
Session Actions Edit View Help
12: command not found
(kali㉿kali)-[~/cyber-log-analysis]
$ grep "Failed password" auth.log
Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2
Nov 06 10:00:08 kali sshd[2102]: Failed password for root from 192.168.56.101 port 50112 ssh2
Nov 06 10:00:15 kali sshd[2103]: Failed password for invalid user test from 192.168.56.101 port 50113 ssh2
Nov 06 10:00:22 kali sshd[2104]: Failed password for invalid user guest from 192.168.56.101 port 50114 ssh2
Nov 06 10:00:29 kali sshd[2105]: Failed password for invalid user oracle from 192.168.56.101 port 50115 ssh2
Nov 06 10:00:36 kali sshd[2106]: Failed password for invalid user admin from 192.168.56.101 port 50116 ssh2
Nov 06 10:01:01 kali sshd[2107]: Failed password for invalid user test from 192.168.56.102 port 50201 ssh2
Nov 06 10:01:15 kali sshd[2108]: Failed password for root from 192.168.56.102 port 50202 ssh2
Nov 06 10:03:01 kali sshd[2110]: Failed password for invalid user admin from 192.168.56.104 port 50401 ssh2
Nov 06 10:05:01 kali sshd[2112]: Failed password for root from 192.168.56.104 port 50402 ssh2
(kali㉿kali)-[~/cyber-log-analysis]
$ grep "Accepted password" auth.log
Nov 06 10:02:01 kali sshd[2109]: Accepted password for kali from 192.168.56.103 port 50301 ssh2
Nov 06 10:04:01 kali sshd[2111]: Accepted password for kali from 192.168.56.103 port 50302 ssh2
(kali㉿kali)-[~/cyber-log-analysis]
$ grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}'
192.168.56.101
192.168.56.101
192.168.56.101
192.168.56.101
192.168.56.101
192.168.56.101
192.168.56.102
192.168.56.102
192.168.56.104
192.168.56.104
(kali㉿kali)-[~/cyber-log-analysis]
$ grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort | uniq -c
6 192.168.56.101
2 192.168.56.102
2 192.168.56.104
(kali㉿kali)-[~/cyber-log-analysis]
$ grep "Failed password" auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort | uniq -c | sort -nr
6 192.168.56.101
2 192.168.56.104
2 192.168.56.102
(kali㉿kali)-[~/cyber-log-analysis]
$ head -n 5 auth.log
Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2


kali@kali: ~/cyber-log-analysis
Session Actions Edit View Help
(kali㉿kali)-[~/cyber-log-analysis]
$ head -n 5 auth.log
Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2
Nov 06 10:00:08 kali sshd[2102]: Failed password for root from 192.168.56.101 port 50112 ssh2
Nov 06 10:00:15 kali sshd[2103]: Failed password for invalid user test from 192.168.56.101 port 50113 ssh2
Nov 06 10:00:22 kali sshd[2104]: Failed password for invalid user guest from 192.168.56.101 port 50114 ssh2
Nov 06 10:00:29 kali sshd[2105]: Failed password for invalid user oracle from 192.168.56.101 port 50115 ssh2
(kali㉿kali)-[~/cyber-log-analysis]
$ tail -n 5 auth.log
Nov 06 10:01:15 kali sshd[2108]: Failed password for root from 192.168.56.102 port 50202 ssh2
Nov 06 10:02:01 kali sshd[2109]: Accepted password for kali from 192.168.56.103 port 50301 ssh2
Nov 06 10:03:01 kali sshd[2110]: Failed password for invalid user admin from 192.168.56.104 port 50401 ssh2
Nov 06 10:04:01 kali sshd[2111]: Accepted password for kali from 192.168.56.103 port 50302 ssh2
Nov 06 10:05:01 kali sshd[2112]: Failed password for root from 192.168.56.104 port 50402 ssh2
(kali㉿kali)-[~/cyber-log-analysis]
$ nano failed_logins.sed
(kali㉿kali)-[~/cyber-log-analysis]
$ sed -n -f failed_logins.sed auth.log
Nov 06 10:00:01 kali sshd[2101]: Failed password for invalid user admin from 192.168.56.101 port 50111 ssh2
Nov 06 10:00:08 kali sshd[2102]: Failed password for root from 192.168.56.101 port 50112 ssh2
Nov 06 10:00:15 kali sshd[2103]: Failed password for invalid user test from 192.168.56.101 port 50113 ssh2
Nov 06 10:00:22 kali sshd[2104]: Failed password for invalid user guest from 192.168.56.101 port 50114 ssh2
Nov 06 10:00:29 kali sshd[2105]: Failed password for invalid user oracle from 192.168.56.101 port 50115 ssh2
Nov 06 10:00:36 kali sshd[2106]: Failed password for invalid user admin from 192.168.56.101 port 50116 ssh2
Nov 06 10:01:01 kali sshd[2107]: Failed password for invalid user test from 192.168.56.102 port 50201 ssh2
Nov 06 10:01:15 kali sshd[2108]: Failed password for root from 192.168.56.102 port 50202 ssh2
Nov 06 10:03:01 kali sshd[2110]: Failed password for invalid user admin from 192.168.56.104 port 50401 ssh2
Nov 06 10:05:01 kali sshd[2112]: Failed password for root from 192.168.56.104 port 50402 ssh2
(kali㉿kali)-[~/cyber-log-analysis]
$ nano analyze.awk
(kali㉿kali)-[~/cyber-log-analysis]
$ sed -n -f failed_logins.sed auth.log | awk -f analyze.awk
FAILED LOGIN ANALYSIS
192.168.56.101 6 attempts SUSPICIOUS
192.168.56.102 2 attempts NORMAL
192.168.56.104 2 attempts NORMAL
(kali㉿kali)-[~/cyber-log-analysis]
$ nano analyze_logs.sh
(kali㉿kali)-[~/cyber-log-analysis]
$ chmod +x analyze_logs.sh


kali@kali: ~/cyber-log-analysis
Session Actions Edit View Help
└─$ nano analyze.awk
┌──(kali㉿kali)-[~/cyber-log-analysis]
└─$ sed -n -f failed_logins.sed auth.log | awk -f analyze.awk
FAILED LOGIN ANALYSIS
192.168.56.101 6 attempts SUSPICIOUS
192.168.56.102 2 attempts NORMAL
192.168.56.104 2 attempts NORMAL
┌──(kali㉿kali)-[~/cyber-log-analysis]
└─$ nano analyze_logs.sh
┌──(kali㉿kali)-[~/cyber-log-analysis]
└─$ chmod +x analyze_logs.sh
┌──(kali㉿kali)-[~/cyber-log-analysis]
└─$ ./analyze_logs.sh
CYBERSECURITY LOG ANALYSIS
Total log records: 12
Failed login attempts: 10
Successful login attempts: 2
FAILED ATTEMPTS BY IP
6 192.168.56.101
2 192.168.56.104
2 192.168.56.102
SUSPICIOUS IP ANALYSIS
FAILED LOGIN ANALYSIS
192.168.56.101 6 attempts SUSPICIOUS
192.168.56.102 2 attempts NORMAL
192.168.56.104 2 attempts NORMAL
SUCCESSFUL LOGIN RECORDS
Nov 06 10:02:01 kali sshd[2109]: Accepted password for kali from 192.168.56.103 port 50301 ssh2
Nov 06 10:04:01 kali sshd[2111]: Accepted password for kali from 192.168.56.103 port 50302 ssh2
Analysis completed.
Reports saved in the output directory.
┌──(kali㉿kali)-[~/cyber-log-analysis]
└─$